THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which method can separate the salt formed after neutralisation from water? 1

(a) filtration **(b)** winnowing **(c)** evaporation **(d)** magnet

2. Why does the colour of a pink China rose solution turn green when washing soda is added? 1

(a) it is a salt **(b)** it is neutral
(c) washing soda is acidic **(d)** washing soda is basic

3. Which is not a use of neutralisation? 1

(a) treating insect bites **(b)** making an acid stronger
(c) treating soil **(d)** curing indigestion

4. Which of the following solutions has a sour taste and corrodes metals? 1

(a) lime water **(b)** sugar solution
(c) soap solution **(d)** dilute hydrochloric acid

5. Which pair would result in release of carbon dioxide? 1

(a) base and salt **(b)** salt and base
(c) water and sugar **(d)** acid and baking soda

6. The product of the reaction between vinegar and baking soda that makes lime water milky is: 1

(a) oxygen **(b)** hydrogen **(c)** carbon dioxide **(d)** nitrogen

7. A student mixes soap solution and lemon juice and the soapy feel disappears. Why? 1

(a) lemon dissolves soap **(b)** acid neutralises the base
(c) soap becomes salt only **(d)** water evaporates

8. Why is it unsafe to taste a base like caustic soda? 1

(a) it is corrosive **(b)** it is sweet **(c)** it is neutral **(d)** it is sticky

9. **Assertion (A):** Baking soda solution is rubbed on an ant sting to soothe it. 1
Reason (R): Baking soda is acidic in nature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Bases such as soap solution feel slippery to touch. 1
Reason (R): Acids such as vinegar taste sour.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do detergents in washing have a basic nature? 2

12. Name the products when sulphuric acid reacts with calcium hydroxide and write the equation. 2

13. Why is lime sprinkled on soil that has become acidic? 2

14. Explain why acidity of curd increases in summer. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does vinegar clean the chalky deposit in a kettle? 3

16. Why do we say that water is neutral? 3

17. Why should we avoid drinking plenty of fizzy drinks regularly? 3

OR

How would you prove that a salt solution is neutral? 3

18. Why should we wear gloves while using floor cleaners that contain strong bases? 3

OR

Describe a method to find which of two shampoos is more basic. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A farmer near Ludhiana finds that wheat grows poorly in one field. A soil test shows that the soil is too acidic. The agriculture officer advises him to spread quicklime (calcium oxide) over the field. A nearby factory also releases acidic waste into a canal, and fish in the canal are dying. 5
(i) What is the nature of the farmer's soil?
(ii) Which substance is added to the soil?
(iii) Name the process by which the soil acidity is reduced.
(iv) What should the factory do before releasing its waste, and why?

20. Explain why neutralisation reactions release heat, and give applications where this is used or must be controlled. 5

OR

Write a short note on the importance of using the right amount of acid or base in everyday products like toothpaste, shampoo and soap. 5

END OF QUESTION PAPER — **Total: 40 Marks**